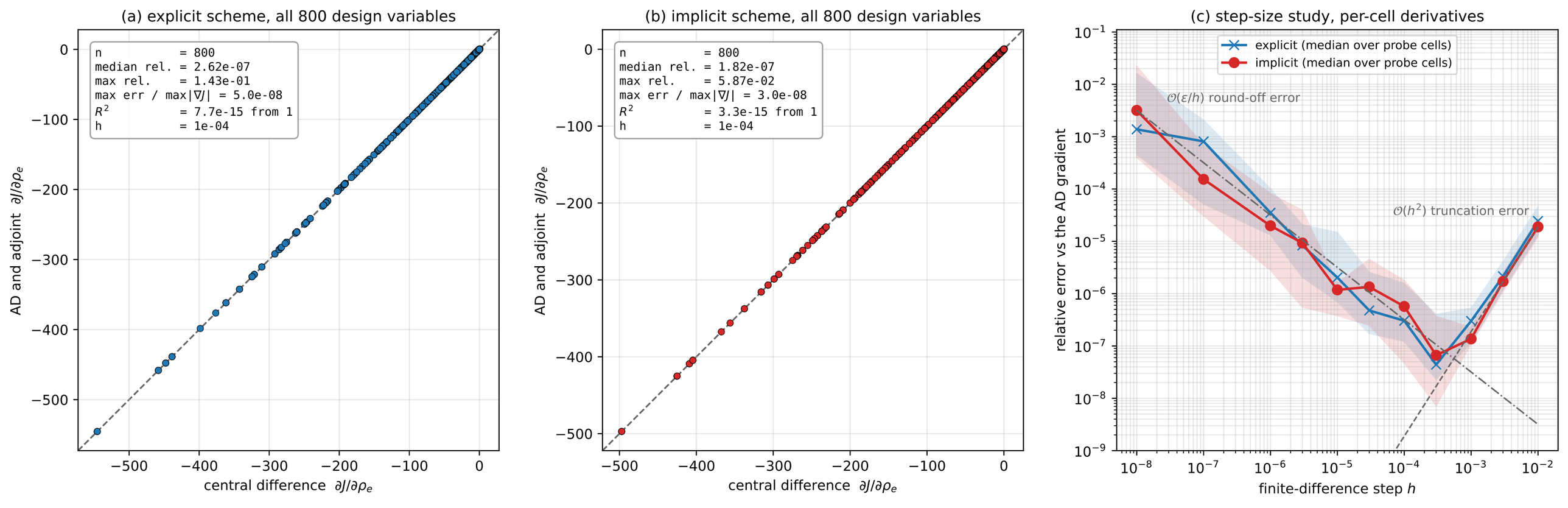

(a) explicit scheme, all 800 design variables
n = 800
median rel. = 2.62e-07
max rel. = 1.43e-01
max err / max|∇J| = 5.0e-08
R^2 = 7.7e-15 from 1
h = 1e-04
AD and adjoint $\partial J/\partial \rho_e$
central difference $\partial J/\partial \rho_e$
(b) implicit scheme, all 800 design variables
n = 800
median rel. = 1.82e-07
max rel. = 5.87e-02
max err / max|∇J| = 3.0e-08
R^2 = 3.3e-15 from 1
h = 1e-04
AD and adjoint $\partial J/\partial \rho_e$
central difference $\partial J/\partial \rho_e$
(c) step-size study, per-cell derivatives
explicit (median over probe cells)
implicit (median over probe cells)
$\mathcal{O}(\varepsilon/h)$ round-off error
$\mathcal{O}(h^2)$ truncation error
relative error vs the AD gradient
finite-difference step h